Project Report on

CUSTOMER BEHAVIOR ANALYSIS

Analyzing Customer Patterns, Purchasing Behavior & Business Insights

PROJECT REPORT

Submitted as part of the **Data Analytics Internship Program** at **Alfido Tech**

Submitted By

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DECLARATION

I hereby declare that the project report entitled **“Customer Behavior Analysis”** is an original work carried out by me as part of my **Data Analyst Internship** at **Alfido Tech**. The project focuses on analyzing customer transaction and behavioral data to identify **customer segments, purchase patterns, and churn risks** using data analytics techniques.

I confirm that the analysis presented in this report was performed by me using **Python, Pandas, NumPy, Matplotlib, Seaborn, SQL, Scikit-learn, and Power BI**. The project involved data cleaning, data preprocessing, feature engineering, exploratory data analysis, customer segmentation using K-Means clustering, churn analysis, SQL-based business analysis, and development of an interactive Power BI dashboard.

I further declare that the findings, visualizations, interpretations, and recommendations presented in this report are based on the analysis performed on the selected dataset. The report has been prepared for academic and internship purposes and has not been submitted previously, in whole or in part, for any other academic or professional qualification.

I take full responsibility for the authenticity and accuracy of the work presented in this report.

Name: Chintu Kumar
Program: B.Tech Computer Science (AI & Data Science)
Organization: Alfido Tech
Project Title: Customer Behavior Analysis

Date: __________________

Place: __________________

Signature of Student

CERTIFICATE

This is to certify that **Mr. Chintu Kumar**, a student of **B.Tech Computer Science (Artificial Intelligence & Data Science)**, has successfully completed the required learning and skill-development activities in the areas of **Data Analytics, SQL, Python, and Power BI**.

During his learning journey, he successfully completed the **Google Data Analytics Professional Certificate from Coursera**, gaining practical exposure to **data cleaning, exploratory data analysis (EDA), data visualization, and analytical problem-solving**.

He has also demonstrated proficiency in **SQL** by solving **50+ SQL problems on LeetCode**, strengthening his query-writing, database analysis, and analytical thinking skills. In addition, he earned an **SQL certification from HackerRank**, validating his understanding of SQL and database querying concepts.

Furthermore, he completed **Power BI micro-course training**, gaining practical knowledge of dashboard development, data visualization, KPI analysis, and interactive reporting. He also completed **Python for Data Science from Infosys Springboard**, developing hands-on skills in Python and commonly used data science libraries such as **Pandas and NumPy**.

Through these certifications, practical exercises, and projects, he has developed a strong foundation in **data analysis, data visualization, SQL, Python, and business intelligence**. His efforts demonstrate a sincere commitment to developing the technical and analytical skills required for a career in the field of **Data Analytics**.

This certificate is issued in recognition of his successful completion of the above-mentioned learning and skill-development activities.

Name: Chintu Kumar
Program: B.Tech Computer Science (Artificial Intelligence & Data Science)

Date: ___________________________

Place: __________________________

Signature: ___________________________

Authorized Signatory / Mentor

Institution / Organization: ___________________________

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Alfido Tech** for providing me with the opportunity to undertake the **Data Analyst Internship** and complete the project entitled **"Customer Behavior Analysis."** This internship has been a valuable learning experience that allowed me to apply my academic knowledge and technical skills to a practical data analytics project.

I am thankful to the management, mentors, supervisors, and team members at **Alfido Tech** for their valuable guidance, support, and feedback throughout the internship. Their guidance helped me understand the practical application of data analytics and improved my approach toward solving business problems using data.

During this project, I gained practical experience in **data cleaning, preprocessing, feature engineering, exploratory data analysis, SQL analysis, customer segmentation, churn analysis, and data visualization**. I worked with tools and technologies including **Python, Pandas, NumPy, Matplotlib, Seaborn, SQL, Scikit-learn, and Power BI**. Developing an interactive Power BI dashboard also helped me understand how analytical results can be presented effectively to support business decision-making.

I would also like to thank my **college faculty members and academic mentors** for providing me with the fundamental knowledge and technical foundation required to complete this project. I am grateful to my **family, friends, and classmates** for their continuous encouragement and support throughout the internship.

This internship enhanced my analytical, technical, problem-solving, and communication skills. It also gave me a better understanding of the complete data analytics process, from preparing raw data to generating meaningful insights and actionable recommendations.

Finally, I sincerely thank everyone who contributed directly or indirectly to the successful completion of this project. The knowledge and practical experience gained during this internship will be valuable for my future academic and professional career in **Data Analytics and Data Science**.

Thank You.

Name: hintu Kumar
Program: B.Tech Computer Science (AI & Data Science)
Organization: Alfido Tech
Project: Customer Behavior Analysis

Date: ____________________
Place: ____________________

Signature: ____________________

ABSTRACT

Customer behavior analysis plays an important role in understanding customer purchasing patterns, preferences, segmentation, and churn risks. This project, **“Customer Behavior Analysis,”** focuses on analyzing customer transaction data to generate meaningful business insights and support data-driven decision-making.

The project follows an end-to-end data analytics approach, beginning with **data cleaning and preprocessing** to identify missing values, duplicates, inconsistent data types, and other data-quality issues. **Feature engineering** was then performed to create useful variables such as purchase month, age group, spending level, and other behavioral attributes. **Exploratory Data Analysis (EDA)** was conducted using Python libraries including **Pandas, NumPy, Matplotlib, and Seaborn** to identify revenue trends, customer behavior, product performance, and purchasing patterns.

SQL was used to perform business-oriented analysis, including revenue analysis, customer behavior analysis, and churn-related queries. **K-Means clustering** was applied for customer segmentation to group customers based on similarities in their purchasing behavior and spending patterns. **Churn analysis** was performed to identify customer groups that may require targeted retention strategies.

An interactive **Power BI dashboard** was developed to present key performance indicators, monthly revenue trends, product category performance, customer segments, spending behavior, and churn risks. The analysis resulted in **8–10 key business findings and 5 actionable recommendations** focused on improving customer retention, increasing customer engagement, and supporting revenue growth.

Overall, the project demonstrates the practical application of **Python, SQL, machine learning, and Power BI** in transforming raw customer data into actionable business insights. It provides an end-to-end understanding of the data analytics process, from data preparation and analysis to visualization, interpretation, and business recommendations.

CHAPTER 1

1.1 INTRODUCTION:

In today's competitive business environment, customer data plays an important role in understanding business performance and making informed decisions. Organizations collect large amounts of data through customer purchases, transactions, product categories, payment methods, demographics, and other activities. However, raw data cannot directly provide meaningful information. It needs to be cleaned, processed, analyzed, and visualized to identify useful patterns and trends. **Customer Behavior Analysis** helps businesses understand customer preferences, purchasing habits, spending patterns, and potential churn risks.

The project titled **"Customer Behavior Analysis"** focuses on analyzing customer transaction and behavioral data to identify **customer segments, purchase patterns, revenue trends, and churn risks**. The dataset contains important attributes such as Customer ID, Purchase Date, Product Category, Product Price, Quantity, Total Purchase Amount, Payment Method, Customer Age, Returns, Gender, and Churn. These attributes are analyzed to understand customer behavior and identify factors that can influence business performance.

The project follows an **end-to-end data analytics process**. The first stage involves **data cleaning and preprocessing**, where missing values, duplicate records, incorrect data types, inconsistent values, and outliers are identified and handled. After cleaning, **feature engineering** is performed to create useful features such as purchase month, purchase year, age group, and spending level. These features help in performing deeper customer and transaction analysis.

Exploratory Data Analysis (EDA) is then performed using **Python, Pandas, NumPy, Matplotlib, and Seaborn**. Different charts and statistical techniques are used to analyze revenue trends, product categories, customer demographics, payment methods, spending behavior, and purchasing patterns. **SQL** is also used to perform business-oriented queries related to customers, revenue, transactions, and churn.

For customer segmentation, **K-Means Clustering** is used to group customers with similar purchasing and spending behavior. This helps identify different customer segments and provides a basis for developing targeted marketing and customer-retention strategies. **Churn Analysis** is also performed to understand the proportion and characteristics of customers who may be lost, helping businesses identify opportunities to improve customer retention.

The final analytical results are presented through an interactive **Power BI dashboard** containing important KPIs such as **Total Revenue, Total Customers, Total Orders, Average Purchase Amount, Churn Rate, and Total Quantity Sold**. The dashboard also presents revenue trends, product performance, customer segments, spending behavior, and churn-related insights in an easy-to-understand visual format.

Overall, this project demonstrates the practical application of **Python, SQL, Machine Learning, and Power BI** to transform raw customer data into meaningful business insights. The analysis supports data-driven decision-making and provides recommendations for improving **customer engagement, retention, marketing effectiveness, and revenue performance**.

1.2 PROBLEM STATEMENT

In today's competitive business environment, organizations collect large amounts of customer transaction data through purchases, product categories, payment methods, customer demographics, and other activities. Although this data contains valuable information, raw data is difficult to interpret without proper analysis. Businesses need an effective analytical approach to understand customer purchasing behavior, identify valuable customer groups, monitor revenue patterns, and detect customers who may be at risk of churn.

The major problem addressed by this project is the **lack of clear and structured insights from raw customer transaction data**. Without proper analysis, businesses may find it difficult to understand which customer segments generate the most revenue, which products or categories perform better, how customer spending varies, and where customer retention efforts are required. This can lead to ineffective marketing strategies and missed opportunities for improving customer satisfaction and revenue.

Therefore, this project aims to analyze customer transaction and behavioral data using **Python, SQL, Machine Learning, and Power BI**. The analysis includes data cleaning, feature engineering, exploratory data analysis, customer segmentation, and churn analysis. The project also focuses on presenting the results through an interactive dashboard and generating actionable recommendations that can support data-driven business decisions.

1.3 OBJECTIVES

The primary objective of the **Customer Behavior Analysis** project is to transform raw customer transaction data into meaningful and actionable insights that can help businesses understand their customers and improve decision-making.

The major objectives of the project are:

- To **clean and preprocess customer data** by identifying and handling missing values, duplicate records, incorrect data types, inconsistent values, and outliers.
- To perform **feature engineering** and create useful attributes from existing customer and transaction data.
- To conduct **Exploratory Data Analysis (EDA)** to understand revenue trends, purchasing patterns, product performance, customer demographics, and spending behavior.
- To use **SQL queries** for structured business analysis and answer important questions related to customers, transactions, revenue, and churn.
- To apply **K-Means Clustering** to divide customers into meaningful groups based on their purchasing and spending behavior.
- To perform **churn analysis** and identify customer groups that may require additional retention and engagement efforts.
- To develop an interactive **Power BI dashboard** containing KPIs, charts, filters, and customer behavior insights.
- To identify **8–10 important business findings** from the overall analysis.
- To provide **actionable recommendations** that can help improve customer retention, customer engagement, marketing effectiveness, and revenue performance.
- To demonstrate a complete **end-to-end data analytics workflow**, from raw data preparation to visualization and business decision-making.
- The overall objective is not only to analyze the dataset but also to understand **how data-driven insights can be converted into practical business strategies**.

1.4 PROJECT GOAL

The primary goal of the **Customer Behavior Analysis** project is to analyze customer transaction and behavioral data and transform it into **meaningful, reliable, and actionable business insights**. The project aims to understand how customers interact with the business, identify different purchasing patterns, recognize valuable customer segments, and determine potential churn risks. By using a systematic data analytics approach, the project seeks to support better and more informed business decision-making.

A major goal of the project is to establish a complete **end-to-end data analytics workflow**. The process begins with understanding and preparing the raw dataset. Data cleaning techniques are applied to handle missing values, duplicate records, incorrect data types, inconsistent values, and potential outliers. This ensures that the dataset is suitable for further analysis and that the results are as accurate and reliable as possible.

Another important goal is to identify meaningful patterns in customer transactions. Through **Exploratory Data Analysis (EDA)**, the project examines revenue trends, purchasing behavior, product categories, customer demographics, payment methods, and spending patterns. These analyses help determine how customer behavior changes across different groups and time periods.

The project also aims to perform **customer segmentation** using **K-Means Clustering**. Customers with similar purchasing and spending characteristics are grouped into different segments. Understanding these segments can help businesses develop targeted strategies, such as loyalty programs for high-value customers, promotional offers for developing customers, and re-engagement campaigns for less-active customers.

A further goal is to analyze **customer churn** and identify patterns associated with customer loss. Understanding churn is important because retaining existing customers can contribute significantly to long-term business performance. The project therefore examines churn across different customer groups and identifies areas where customer retention strategies may be required.

SQL is used as an additional analytical tool to answer business-related questions and extract useful information from customer data. The project aims to demonstrate how SQL can be used for revenue analysis, customer analysis, transaction analysis, and churn-related queries.

Another key goal is to develop an interactive **Power BI dashboard** that presents the most important results in a clear and visually understandable format. The dashboard includes KPIs such as **Total Revenue, Total Customers, Total Orders, Average Purchase Amount, Churn Rate, and Total Quantity Sold**, along with charts for revenue trends, customer segments, product performance, spending behavior, and churn.

Finally, the project aims to convert analytical results into **business findings and actionable recommendations**. Rather than simply presenting charts and numerical results, the project focuses on explaining what the findings mean from a business perspective and how they can be used to improve customer retention, engagement, marketing strategies, and revenue performance.

Overall, the goal of this project is to demonstrate how **Python, SQL, Machine Learning, and Power BI** can work together to transform raw customer data into useful business knowledge and support **data-driven decision-making**.

CHAPTER 2

2.1 SOFTWARE REQUIREMENTS

The **Customer Behavior Analysis** project requires software tools for data cleaning, analysis, visualization, customer segmentation, SQL analysis, and dashboard development. **Python** is used as the main programming language, while **Jupyter Notebook** is used for writing and executing the analysis.

The major Python libraries used are **Pandas and NumPy** for data processing, **Matplotlib and Seaborn** for visualization, and **Scikit-learn** for K-Means customer segmentation. **SQL/PostgreSQL** is used for database-based customer and business analysis. **Power BI** is used to create an interactive dashboard containing KPIs, charts, filters, and business insights.

Software Used

Software	**Purpose**
Windows 10/11	Operating System
Python	Data Analysis
Jupyter Notebook	Coding & Analysis
Pandas & NumPy	Data Processing
Matplotlib & Seaborn	Visualization
Scikit-learn	Customer Segmentation
PostgreSQL / SQL	Business Analysis
Power BI	Dashboard Development

These tools provide a complete environment for performing **data cleaning, EDA, SQL analysis, customer segmentation, churn analysis, and business reporting**.

2.2 HARDWARE REQUIREMENTS

The project can be completed using a standard modern computer because the customer dataset does not require high-end computing resources. The system should have sufficient processing power, memory, and storage to run Python, Jupyter Notebook, PostgreSQL, and Power BI efficiently.

Adequate storage is required for datasets, software, notebooks, Power BI files, and project documentation. An **SSD** is preferred because it provides faster application loading and data processing.

Hardware	**Requirement**
Processor	Intel Core i3 / Ryzen 3 or above
RAM	8 GB minimum
Storage	10 GB+ available
Display	HD / Full HD
Operating System	Windows 10/11 64-bit
Internet	Stable Internet Connection
Keyboard & Mouse	Standard

2.3 TOOLS & TECHNOLOGIES

The **Customer Behavior Analysis** project uses a combination of programming, database, machine learning, and business intelligence technologies to perform complete data analysis. Each technology is used for a specific stage of the project, from data preparation to final visualization and reporting.

Python is used as the primary programming language for data analysis. It provides a flexible environment for cleaning, transforming, and analyzing customer transaction data. **Pandas** is used for data manipulation and preprocessing, while **NumPy** supports numerical calculations and statistical operations.

For **Exploratory Data Analysis (EDA)**, **Matplotlib** and **Seaborn** are used to create graphs and visualizations. These libraries help identify revenue trends, purchasing patterns, customer demographics, product performance, and relationships between numerical variables.

SQL/PostgreSQL is used for structured database analysis. SQL queries are performed to analyze customer transactions, revenue, product categories, customer behavior, and churn-related information. It helps answer practical business questions using filtering, grouping, aggregation, and other SQL operations.

Scikit-learn is used for applying **K-Means Clustering** to segment customers according to their purchasing and spending behavior. Customer segmentation helps identify groups of customers with similar characteristics and supports targeted business strategies.

Microsoft Power BI is used for developing the final interactive dashboard. It presents important KPIs such as **Total Revenue, Total Customers, Total Orders, Average Purchase Amount, and Churn Rate**, along with charts and filters that make the analysis easy to understand.

Tools & Technologies Used

Tool / Technology	**Purpose**
Python	Data Analysis & Processing
Jupyter Notebook	Development & Documentation
Pandas	Data Cleaning & Manipulation
NumPy	Numerical Analysis
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
SQL / PostgreSQL	Database & Business Analysis
Scikit-learn	K-Means Customer Segmentation
Power BI	Interactive Dashboard & Reporting

Together, these technologies provide a complete **end-to-end data analytics workflow**, covering data cleaning, feature engineering, EDA, SQL analysis, customer segmentation, churn analysis, visualization, and business insight generation.

CHAPTER 3

3.1 DATASET DESCRIPTION

The dataset used for this project is **"Customer Behavior Analysis"**, obtained from Kaggle. It contains customer transaction, demographic, and behavioral information that can be analyzed to understand **purchasing patterns, spending behavior, customer segments, revenue trends, and churn risks**.

The dataset contains information related to customer identification, purchase details, product categories, transaction amounts, payment methods, customer age, gender, returns, and churn status. Before analysis, the dataset is checked and prepared through data cleaning and preprocessing. This includes checking **missing values, duplicate records, data types, unique values, and outliers**. The Purchase Date column is converted into a proper date format to support monthly and time-based analysis.

The cleaned dataset is then used for **Exploratory Data Analysis (EDA), SQL business analysis, customer segmentation, churn analysis, and Power BI dashboard development**. The dataset provides sufficient information to identify important customer and transaction patterns and generate actionable business insights.

3.2 DATASET FEATURES

The major features available in the dataset are:

Feature	Description
Customer ID	Unique identification of each customer
Purchase Date	Date of the customer transaction
Product Category	Category of the purchased product
Product Price	Price of the purchased product
Quantity	Number of products purchased
Total Purchase Amount	Total amount spent in the transaction
Payment Method	Method used for payment
Customer Age / Age	Age-related customer information
Returns	Information about returned purchases
Customer Name	Customer name
Gender	Gender of the customer
Churn	Indicates the customer's churn status

The dataset contains both **numerical and categorical features**. Numerical variables such as Product Price, Quantity, Total Purchase Amount, and Age are useful for statistical analysis and customer segmentation. Categorical variables such as Product Category, Payment Method, Gender, and Churn are useful for comparing different customer groups.

These features collectively provide the foundation for analyzing **customer behavior, purchasing patterns, revenue performance, segmentation, and churn risks** in the project.

3.3 PROJECT METHODOLOGY

The **Customer Behavior Analysis** project follows an end-to-end data analytics methodology to convert raw customer data into meaningful business insights. The major steps of the methodology are as follows:

1. **Data Collection:**
   The customer behavior dataset is collected from Kaggle and examined to understand its structure, columns, and data types.
2. **Data Cleaning:**
   The dataset is checked for **missing values, duplicate records, incorrect data types, inconsistent values, and outliers**. Required corrections are performed before analysis.
3. **Feature Engineering:**
   Useful features such as **purchase month, purchase year, age group, and spending level** are created from the existing data to improve the analysis.
4. **Exploratory Data Analysis (EDA):**
   Python tools such as **Pandas, NumPy, Matplotlib, and Seaborn** are used to analyze revenue, purchasing patterns, customer demographics, product categories, and spending behavior.
5. **SQL Analysis:**
   SQL/PostgreSQL is used to perform business-related queries for analyzing customers, transactions, revenue, purchasing behavior, and churn.
6. **Customer Segmentation:**
   K-Means Clustering is applied to group customers with similar purchasing and spending behavior into different customer segments.
7. **Churn Analysis:**
   Customer churn is analyzed to identify churn patterns and customer groups that may require retention strategies.
8. **Power BI Dashboard:**
   An interactive **Power BI dashboard** is developed using KPIs, charts, filters, and slicers to present the major analytical results.
9. **Business Insights:**
   Finally, the results are interpreted to identify important **business findings and actionable recommendations** for improving customer retention, engagement, and revenue.

Project Methodology Flow

Data Collection → Data Cleaning → Feature Engineering → EDA → SQL Analysis → Customer Segmentation → Churn Analysis → Power BI Dashboard → Business Insights & Recommendations.

3.4 PROJECT WORKFLOW

The project workflow represents the complete step-by-step process followed to analyze customer behavior and generate business insights. It begins with data collection and ends with dashboard development and actionable recommendations. Each stage is important and contributes to the final analytical results.

The workflow ensures that raw data is transformed into meaningful information through proper cleaning, analysis, segmentation, and visualization. The following diagram shows the overall workflow of the project.

This workflow ensures a structured and effective approach to Customer Behavior Analysis. It helps in understanding customer purchasing patterns, identifying valuable customer segments, analyzing churn, and providing data-driven recommendations. The combination of Python analysis, SQL queries, clustering, and Power BI visualization provides a complete end-to-end analytics solution.

CHAPTER 4

4.1 DATA CLEANING

Data cleaning prepares the dataset for accurate analysis. The data is checked for **missing values, duplicates, incorrect data types, and inconsistent values** using Pandas.

The `Purchase Date` column is converted into the correct date format. After cleaning, the dataset is ready for **EDA, SQL, segmentation, churn analysis, and Power BI**.

4.2 Missing Values & Duplicates

Missing values are checked using `isnull().sum()`, while duplicate records are identified using `duplicated()` and removed using `drop_duplicates()`.

The dataset is checked again after cleaning to ensure **data accuracy and consistency**.

```
•[615]: #Data Cleaning
        df.columns=df.columns.str.lower()
        df.columns=df.columns.str.replace(" ","_")

[616]:  df.isnull().sum()

        customer_id 0 •••

[617]:  df['returns'].value_counts(dropna=False)

        returns •••

[618]:  df['returns']=df['returns'].fillna('No')

[619]:  df.isnull().sum()

[619]:  customer_id            0
        purchase_date          0
        product_category       0
        product_price          0
        quantity               0
        total_purchase_amount  0
        payment_method         0
        customer_age           0
        returns                0
        customer_name          0
        age                    0
        gender                 0
        churn                  0
        dtype: int64

[632]:  df.duplicated().sum()

[632]:  0
```

Figure 4.2: Missi ng Values & Duplicate Check

4.3 OUTLIER DETECTION

Outlier detection was performed on the numerical columns of the dataset, including **Product Price, Quantity, Total Purchase Amount, and Age**. The data was examined using statistical methods and box plots to identify unusually high or low values.

The analysis showed that **no significant outliers were detected in the dataset**. Therefore, no records were removed or modified during this step. The original values were retained to preserve the accuracy and completeness of the customer transaction data.

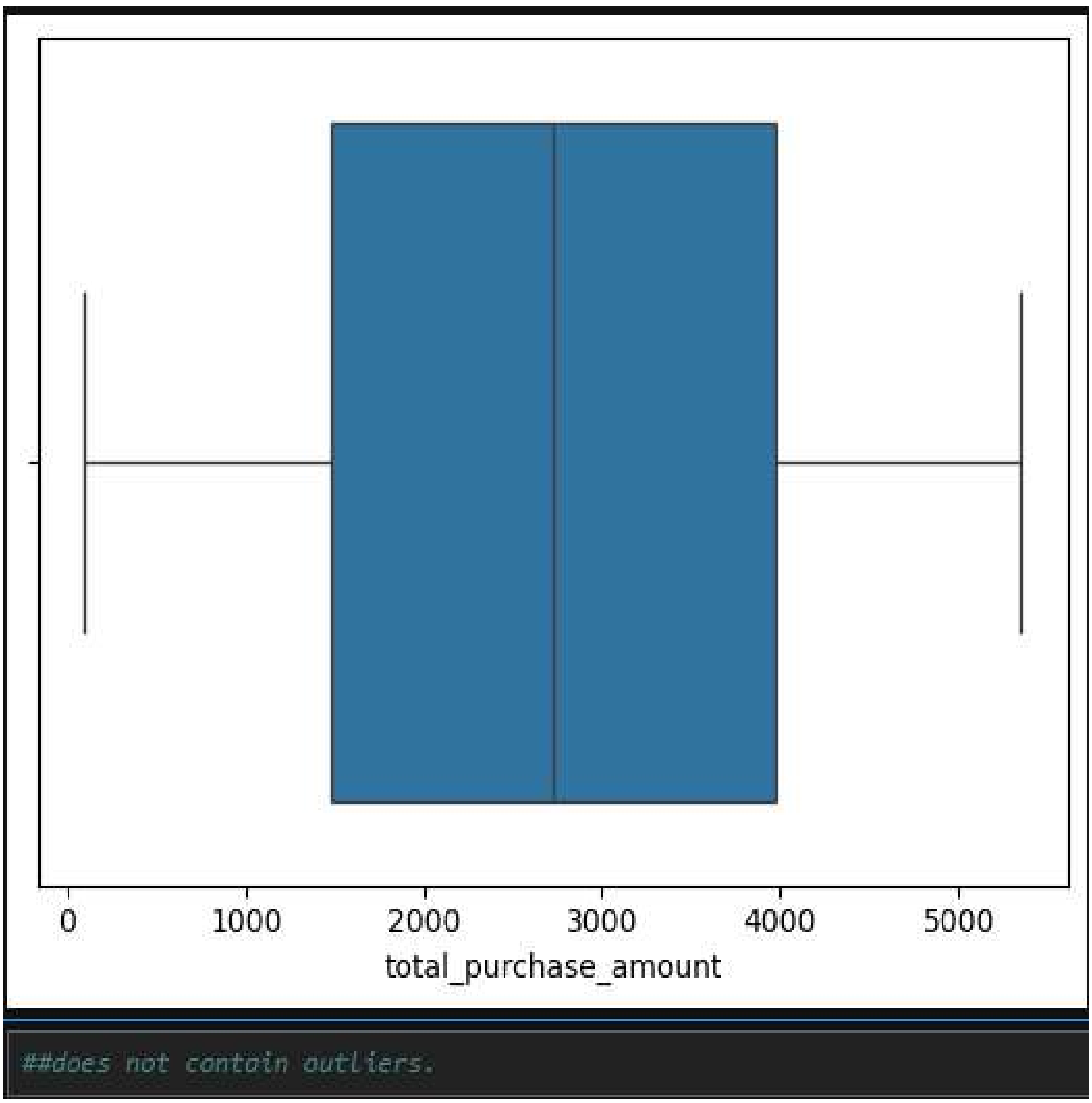


Figure 4.3: Outlier Detection Result – No Significant Outliers Found

4.4 FEATURE ENGINEERING

In this project, **feature engineering** is performed to create useful new features from the existing customer transaction data. These features help in analyzing **customer behavior, purchase patterns, spending levels, and product characteristics**.

The following features were created:

- **Purchase Year** – Extracted from `purchase_date` to perform yearly analysis.
- **Purchase Month** – Extracted from `purchase_date` to analyze monthly purchase trends.
- **Purchase Day** – Extracted from `purchase_date` to analyze daily purchasing behavior.
- **Purchase Quarter** – Extracted from `purchase_date` to perform quarterly analysis.
- **Age Group** – Customers are divided into groups such as **18–25, 26–35, 36–45, 46–60, and 60+**.
- **Spending Level** – Customers are classified as **Low, Medium, or High** based on their total purchase amount.
- **Price Category** – Products are categorized as **Low Price, Medium Price, or High Price**.
- **Quantity Category** – Purchase quantity is categorized as **Low, Medium, or High**.

These engineered features make the dataset more suitable for **EDA, SQL analysis, customer segmentation, churn analysis, and Power BI visualization**.

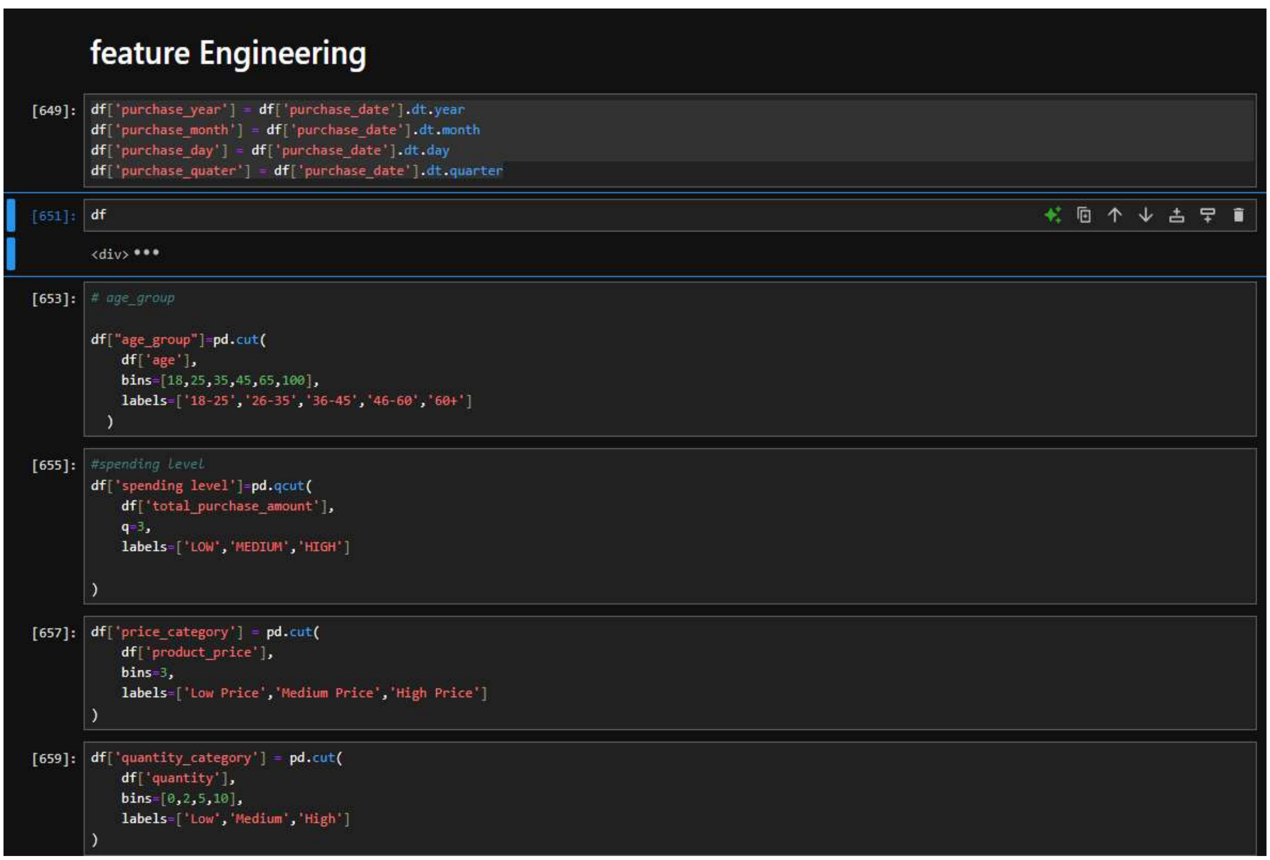


Figure 4.4: Feature Engineering – Creation of New Features

5.1 REVENUE ANALYSIS

Revenue analysis is an important part of the **Customer Behavior Analysis** project. It helps evaluate the company's overall sales performance and understand how revenue changes across **time, product categories, and customer segments**. The analysis was performed using **Power BI** through KPI cards, bar charts, and line charts.

The dashboard shows a **Total Revenue of approximately 681M** and an **Average Purchase of approximately 2.73K**. These KPIs provide a quick overview of the overall financial performance of the business. The revenue trend by purchase month shows fluctuations throughout the year. Revenue remains comparatively high during the initial months, while a noticeable decline can be seen in the later months, followed by a small recovery. This pattern indicates changes in customer purchasing activity over time and can help the business identify periods requiring additional promotional activities.

The **product category analysis** compares revenue generated by different categories such as **Home, Clothing, Electronics, and Books**. The dashboard indicates that all four categories contribute significantly to total revenue. Home and Clothing show strong performance, while Electronics and Books also make a considerable contribution. This analysis can help the business understand which product categories should receive greater attention in terms of inventory, marketing, and promotional campaigns.

Customer-segment analysis provides another important view of revenue performance. **Customer Segment 1 and Segment 2** generate the highest revenue, contributing approximately **244M and 243M**, respectively. In comparison, **Segment 0 and Segment 3** contribute approximately **99M and 96M**. Therefore, Segments 1 and 2 can be considered important customer groups because of their higher contribution to overall revenue.

The dashboard also provides interactive filters for **Gender, Payment Method, Purchase Month, Product Category, and Customer Segment**. These filters allow users to examine revenue for specific customer groups and purchasing conditions. For example, revenue can be compared between male and female customers or analyzed for a particular product category and month.

Revenue analysis can support business decisions such as identifying high-performing products, focusing on valuable customer segments, planning promotional campaigns, and monitoring changes in monthly sales. Combining revenue analysis with customer spending and churn analysis can provide a more complete understanding of customer behavior.

Key Findings

- Total Revenue is approximately **681M**.
- Average Purchase Amount is approximately **2.73K**.
- Revenue shows fluctuations across the purchase months.
- A significant revenue decline is visible during the later months.
- **Home and Clothing** are among the strong revenue-generating categories.
- **Segments 1 and 2** contribute the highest revenue.
- Interactive filters allow detailed revenue analysis by customer characteristics.
- Revenue trends can help identify opportunities for **sales growth and customer engagement**.

5.2 CUSTOMER & PRODUCT ANALYSIS

Customer and product analysis is performed to understand **customer distribution, customer segments, order volume, and product-wise revenue contribution**. The dashboard shows approximately **50K customers and 250K orders**, indicating a large volume of customer transactions.

The customer segmentation analysis shows four segments: **0, 1, 2, and 3**. Segments **1 and 2** generate the highest revenue, approximately **244M and 243M**, while Segments 0 and 3 contribute comparatively less. This indicates that Segments 1 and 2 are important customer groups and can be given greater attention through loyalty programs and targeted offers.

Product analysis includes **Home, Clothing, Electronics, and Books**. All categories contribute significantly to total revenue, with **Home and Clothing** showing strong performance. Comparing categories helps the business identify products that require greater marketing attention and better inventory planning.

Overall, customer and product analysis provides useful information about **valuable customer groups and high-performing product categories**, supporting better marketing, inventory, and customer-retention decisions.

5.3 Purchase & Spending Analysis

Purchase and spending analysis focuses on **order activity, average purchase value, monthly purchasing patterns, and customer spending levels**. The dashboard records approximately **250K orders** and an **Average Purchase Amount of 2.73K**, providing an overview of customer transaction behavior.

The monthly revenue trend shows fluctuations in purchasing activity. Revenue is comparatively stronger during the earlier months and shows a noticeable decline during the later months. This may indicate changes in customer purchasing activity and can help the business identify periods where promotional campaigns may be required.

Customers are classified into **Low, Medium, and High spending levels**. This classification helps identify customers who contribute more to revenue and provides a basis for targeted marketing strategies. High-spending customers can be encouraged through loyalty benefits, while medium- and low-spending customers can be targeted with suitable offers and cross-selling strategies.

The dashboard also allows purchase behavior to be analyzed using **Gender, Payment Method, Purchase Month, Product Category, and Customer Segment**. These filters make it possible to compare different customer groups and identify specific purchasing patterns.

Overall, purchase and spending analysis helps the business understand **how customers purchase, how much they spend, and which customer groups provide greater value**. These insights can support customer engagement, retention, promotional planning, and revenue growth.

5.4 EDA FINDING

Exploratory Data Analysis (EDA) was performed using **Python and Pandas** to understand the overall characteristics and purchasing behavior of customers. Key Performance Indicators (KPIs), product categories, payment methods, and gender distribution were analyzed.

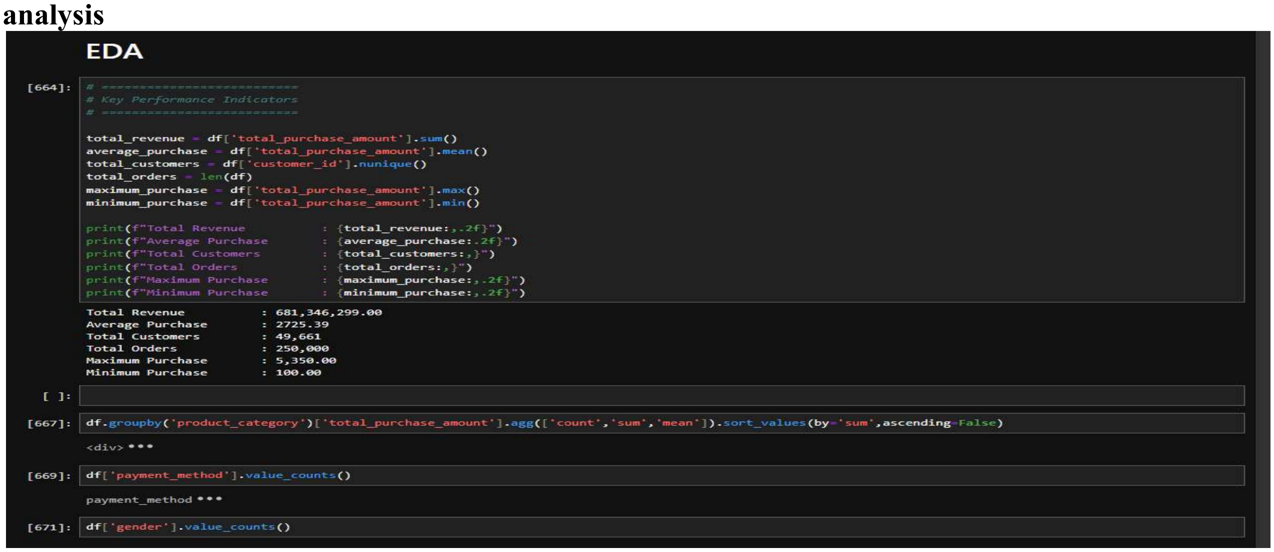
The analysis shows that the dataset contains **250,000 customer transactions** and **49,661 unique customers**. The **total purchase revenue is ₹681,346,299**, while the **average purchase amount is ₹2,725.39**. The minimum purchase amount is **₹100**, and the maximum purchase amount is **₹5,350**, showing variation in customer spending.Product-wise analysis was performed using `groupby()` to compare the **count, total purchase amount, and average purchase amount** across different product categories.

The analysis also examined **payment methods** using `value_counts()` to understand customer preferences for different payment options. Similarly, **gender distribution** was analyzed to understand the composition of the customer base and compare purchasing behavior between customer groups.

Key EDA Findings

- **Total Revenue:** ₹681.35M
- **Total Customers:** 49,661
- **Total Orders:** 250,000
- **Average Purchase:** ₹2,725.39
- **Maximum Purchase:** ₹5,350
- **Minimum Purchase:** ₹100
- Product categories were compared based on **revenue, transaction count, and average purchase**.
- Payment methods were analyzed to identify **customer payment preferences**.
- Gender distribution was examined to understand the **customer composition**.

These findings provide a foundation for further **customer segmentation, spending analysis, churn analysis**



EDA

```
[664]: # ===========================
       # Key Performance Indicators
       # ===========================

       total_revenue = df['total_purchase_amount'].sum()
       average_purchase = df['total_purchase_amount'].mean()
       total_customers = df['customer_id'].nunique()
       total_orders = len(df)
       maximum_purchase = df['total_purchase_amount'].max()
       minimum_purchase = df['total_purchase_amount'].min()

       print(f"Total Revenue          : {total_revenue:,.2f}")
       print(f"Average Purchase       : {average_purchase:.2f}")
       print(f"Total Customers        : {total_customers:,}")
       print(f"Total Orders           : {total_orders:,}")
       print(f"Maximum Purchase       : {maximum_purchase:,.2f}")
       print(f"Minimum Purchase       : {minimum_purchase:,.2f}")

       Total Revenue        : 681,346,299.00
       Average Purchase     : 2725.39
       Total Customers      : 49,661
       Total Orders         : 250,000
       Maximum Purchase     : 5,350.00
       Minimum Purchase     : 100.00

[ ]:

[667]: df.groupby('product_category')['total_purchase_amount'].agg(['count','sum','mean']).sort_values(by='sum',ascending=False)

       <div> •••

[669]: df['payment_method'].value_counts()

       payment_method •••

[671]: df['gender'].value_counts()
```

CHAPTER 6

6.1 SQL ANALYSIS

SQL analysis was performed using **MySQL** to extract meaningful business insights from the cleaned customer transaction dataset. SQL queries were used to filter, group, sort, and aggregate the data. Functions such as **COUNT(), SUM(), AVG(), GROUP BY, ORDER BY, WHERE, HAVING, and CASE** were used to analyze customer and transaction behavior.

The analysis focused on important business questions such as total revenue, customer count, average purchase amount, product performance, customer spending, and churn. SQL provided a structured way to analyze large numbers of transactions and validate the results used in the Power BI dashboard.

6.2 Customer & Revenue Analysis

Customer and revenue analysis was conducted to identify **high-value customers, revenue-generating products, and customer purchasing patterns**. Customers were analyzed based on their purchase amounts, order frequency, product categories, and customer segments.

The analysis showed total revenue of approximately **₹681.35M**, with an average purchase amount of approximately **₹2,725.39**. Product-wise revenue was compared to identify categories contributing more to overall sales. Customer segments were also analyzed to determine which groups generated higher revenue.

SQL analysis helped convert raw transaction data into **business-oriented insights** that can support customer targeting, revenue optimization, product planning, and retention strategies. The results were further used to support the findings presented in the **Power BI dashboard**.

Revenue by Product Category

	product_category text	revenue numeric
1	Home	171138916
2	Clothing	170716122
3	Electronics	170146025
4	Books	169345236

6.3 CHURN ANALYSIS

Churn analysis was performed to identify customers who have **stopped purchasing or are at risk of leaving**. The dataset contains a `churn` variable, where **0 represents customers who have not churned and 1 represents churned customers**.

SQL was used to analyze churn by **customer segment, age group, gender, and spending level**. This helped identify customer groups with higher churn and compare their purchasing behavior with retained customers.

The analysis was also connected with the Power BI dashboard, where churn is visualized across **customer segments and age groups**. The overall churn rate shown in the dashboard is approximately **20%**, indicating that a portion of customers may require additional engagement.

-- Overall Churn Count

	churn bigint	total_customers bigint
1	0	199870
2	1	50130

--Churn Percentage

	churn bigint	total_customers bigint	churn_percentage numeric
1	0	199870	79.95
2	1	50130	20.05

--3. Churn by Age_Group

	age_group text	churn bigint	total_customers bigint
1	18-25	0	26896
2	18-25	1	7294
3	26-35	0	38359
4	26-35	1	9255
5	36-45	0	37545
6	36-45	1	9280
7	46-60	0	74453
8	46-60	1	18547
9	60+	0	18744
10	60+	1	4839
11	[null]	0	3873
12	[null]	1	915

CHAPTER 7

7.1 CUSTOMER SEGMENTATION

Customer segmentation was performed to divide customers into different groups based on their **purchasing behavior and spending patterns**. In this project, customers were categorized into four customer segments: **0, 1, 2, and 3**. Segmentation helps the business understand differences between customer groups and identify high-value customers.

The dashboard shows that **Segments 1 and 2** are the strongest revenue-generating groups, contributing approximately **₹244M and ₹243M**, respectively. Segments 0 and 3 contribute approximately **₹99M and ₹96M**. This indicates that Segments 1 and 2 are more valuable in terms of revenue contribution.

Customers were also classified into **Low, Medium, and High spending levels**. This provides an additional view of customer value and helps identify customers who spend more or less. High-spending customers can be targeted with loyalty benefits, personalized offers, and premium promotions, while low-spending customers can be encouraged through discounts and cross-selling strategies.

The segmentation analysis also considers **age group, gender, purchase behavior, and churn status**. Combining these characteristics provides a better understanding of customer profiles and helps identify groups that may require different marketing strategies.

Overall, customer segmentation helps the business move from general marketing to **targeted customer engagement**. It supports better decisions related to customer retention, personalized marketing, promotions, and revenue growth.


```
#You can describe the segments like this (adapt to your actual results):
#Segment Description of Above graph
#0  Young, low-spending customers
#1  High-value customers with high purchase amounts
#2  Medium-spending regular customers
#3  Older customers with high quantities purchased
```

7.2 K-MEAN CLUSTERING

K-Means Clustering is an **unsupervised machine learning technique** used to divide customers into groups based on similarities in their purchasing behavior. In the Customer Behavior Analysis project, K-Means can be used to identify customer groups with similar **spending and purchasing patterns**.

For clustering, relevant customer-level features such as **total purchase amount, purchase frequency, and average purchase value** can be considered. Before applying K-Means, the selected numerical features are standardized so that variables with larger values do not dominate the clustering process.

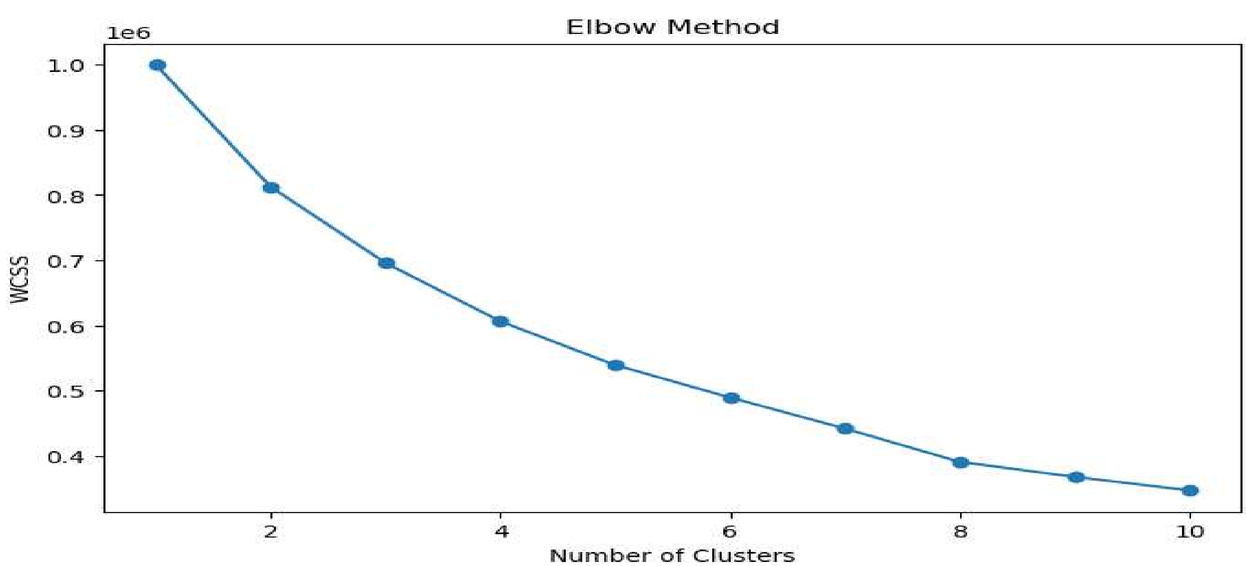
The K-Means algorithm requires the number of clusters (**K**) to be selected. The **Elbow Method** can be used to compare different values of K and identify a suitable number of customer groups. After selecting K, the algorithm assigns each customer to the nearest cluster based on the distance between their feature values. The cluster centers are then updated repeatedly until the groups become stable.

In this project, clustering can help identify different customer profiles such as **low-value, medium-value, and high-value customers**. These groups can be compared with the existing spending levels and customer segments shown in the Power BI dashboard.

Customer Segmentation Through Clustering

Cluster Type	**Customer Behavior**	**Possible Strategy**
Low-Value	Lower spending/purchase activity	Discounts and engagement offers
Medium-Value	Moderate spending	Cross-selling and personalized offers
High-Value	Higher spending and purchase value	Loyalty rewards and premium offers

K-Means clustering provides a data-driven approach to customer segmentation. It can help the business understand **customer similarities, identify valuable customers, and develop targeted marketing strategies**. The clustering results can also be compared with the existing customer segments to improve the overall customer behavior analysis.



CHAPTER 8

8.1 DASHBOARD & KPIs

A **Power BI dashboard** was developed to present customer behavior and business performance in an interactive format. The dashboard contains six major KPIs:

- **Total Revenue:** 681M
- **Total Customers:** 50K
- **Total Orders:** 250K
- **Average Purchase:** 2.73K
- **Churn Rate:** 20%
- **Total Quantity Sold:** 751K

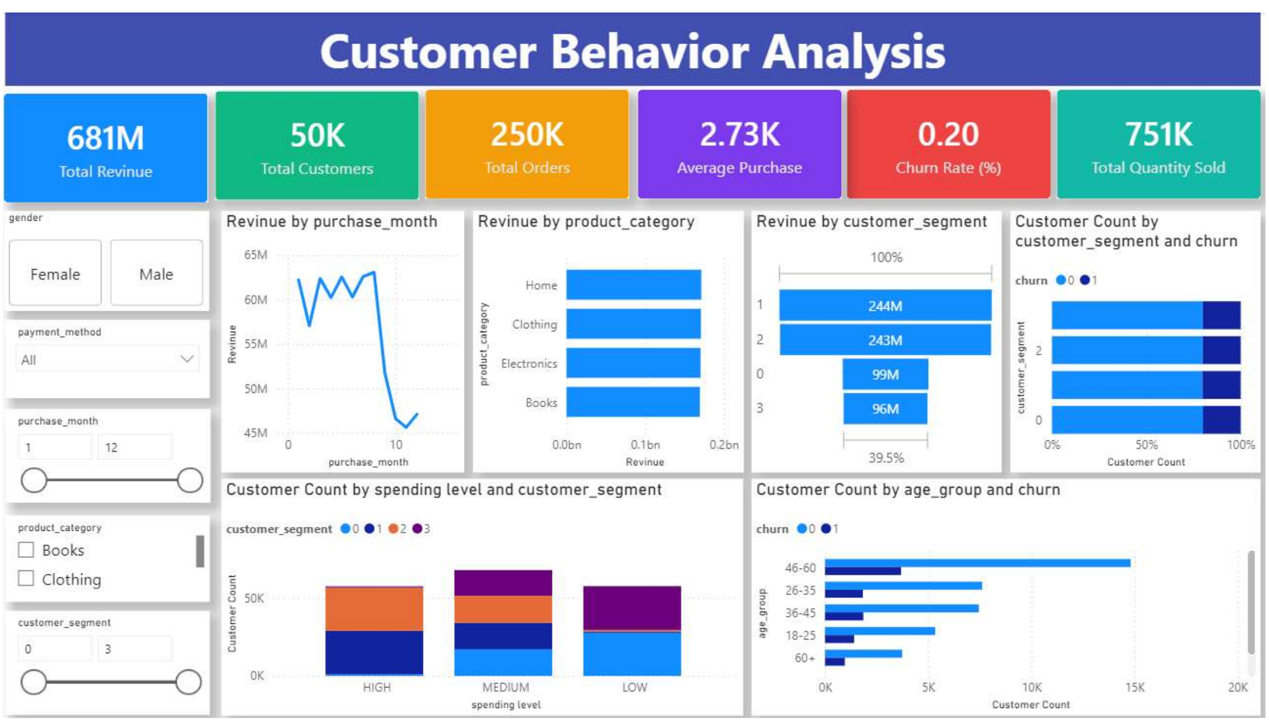
These KPIs provide a quick overview of sales, customers, purchasing activity, and churn.

8.2 Visualizations & Filters

The dashboard includes visualizations for **monthly revenue, product-wise revenue, customer segments, spending levels, and churn by age group**. These charts help identify important customer and sales patterns.

Interactive filters are provided for **Gender, Payment Method, Purchase Month, Product Category, and Customer Segment**. Users can apply these filters to analyze specific customer groups and obtain detailed insights.

Overall, the dashboard provides a simple and interactive way to monitor **revenue, customer behavior, spending, product performance, and churn** for better business decision-making.



CHAPTER 9

9.1 BUSINESS FINDING

Based on the **EDA, SQL analysis, and Power BI dashboard**, the following key findings were identified:

1. The dataset contains approximately **50K customers and 250K orders**, generating around **681M revenue**.
2. The **average purchase amount is 2.73K**, indicating significant customer spending.
3. **Customer Segments 1 and 2** generate the highest revenue, approximately **244M and 243M**.
4. **Home and Clothing** are among the strong revenue-generating product categories.
5. Monthly revenue shows fluctuations with a **noticeable decline in later months**.
6. The **46–60 age group** has the highest customer count in the dashboard.
7. The overall **churn rate is approximately 20%**, showing a need for better customer retention.
8. Customers are distributed across **Low, Medium, and High spending levels**, indicating different customer-value groups.
9. **Customer Quantity:** Around **751K units** were sold, indicating a high level of purchasing activity across the customer base.
10. **Payment Behavior:** Different payment methods show varying customer preferences, which can help the business optimize payment options and improve the purchasing experience.

9.2 Actionable Recommendations

1. **Focus on high-value customers:** Provide loyalty rewards and personalized offers to Segments 1 and 2.
2. **Reduce churn:** Target at-risk customers with discounts, loyalty benefits, and personalized communication.
3. **Increase spending:** Use cross-selling, product recommendations, and special offers for Low and Medium-spending customers.
4. **Promote strong categories:** Increase marketing efforts for **Home and Clothing** while monitoring other categories.
5. **Improve weak-month sales:** Introduce seasonal promotions and targeted campaigns during months with lower revenue.
6. **Use customer segmentation:** Create different marketing strategies based on **age, spending level, customer segment, and churn status**.
7. **Age-Based Marketing:** Create targeted campaigns for the **46–60 age group**, while developing different offers for other age groups based on their behavior.
8. **Regular KPI Monitoring:** Continuously monitor **revenue, orders, average purchase, customer count, quantity sold, and churn rate** through the Power BI dashboard to identify changes quickly.

Overall, these recommendations can help improve **customer retention, purchasing activity, and revenue growth**.

CONCLUSION & FUTURE SCOPE

CONCLUSION

The **Customer Behavior Analysis** project successfully analyzed customer transactions to understand **purchasing patterns, customer segments, spending behavior, product performance, and churn risks**. The project used **Python, Pandas, SQL, and Power BI** for data cleaning, analysis, and visualization.

The dataset contains approximately **250K orders and 49,661 customers**, with total revenue of around **681.35M** and an average purchase amount of approximately **2.73K**. Feature engineering was performed by creating **purchase year, month, day, and quarter** from the purchase date. EDA and SQL analysis were then used to identify important business patterns.

The Power BI dashboard presents key KPIs such as **Total Revenue, Total Customers, Total Orders, Average Purchase, Churn Rate, and Total Quantity Sold**. The analysis found that **Customer Segments 1 and 2** generate the highest revenue, while **Home and Clothing** are among the strong-performing product categories. The overall churn rate of approximately **20%** highlights the importance of customer retention.

Overall, the project demonstrates how data analytics can convert raw transaction data into **meaningful business insights** and support better data-driven decisions.

Future Scope

The project can be further improved by developing a **machine learning-based churn prediction model** to identify customers who are likely to leave. Advanced **K-Means clustering** can be applied to create more detailed customer groups based on spending, purchase frequency, and customer value.

Future improvements can include **Customer Lifetime Value (CLV) prediction** to identify high-value customers and **personalized product recommendation systems** based on individual purchasing history. **Sales forecasting** can also be implemented to predict future revenue and support inventory planning.

The Power BI dashboard can be enhanced with **real-time data updates, automated data refresh, drill-through reports, and additional interactive filters**. Integration with a database can allow the dashboard to automatically reflect new transactions.

Further analysis can include **customer retention prediction, product demand forecasting, seasonal analysis, payment behavior analysis, and marketing campaign evaluation**. These improvements can help businesses make faster decisions, improve customer satisfaction, increase retention, and achieve sustainable revenue growth.

BIBLIOGRAPHY / REFERENCES

The following resources were referred to during the development of the **Customer Behavior Analysis** project for understanding data cleaning, exploratory data analysis, SQL queries, Python programming, machine learning concepts, and Power BI visualization.

Books & Learning Resources

1. McKinney, Wes. **Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython.** O'Reilly Media.
2. Han, Jiawei, Kamber, Micheline, and Pei, Jian. **Data Mining: Concepts and Techniques.** Morgan Kaufmann.
3. Kotu, Vijay, and Deshpande, Bala. **Data Science: Concepts and Practice.** Morgan Kaufmann.

Online Documentation

4. **Python Documentation** – Used for understanding Python programming concepts and functions.
5. **Pandas Documentation** – Used for data cleaning, transformation, feature engineering, and exploratory data analysis.
6. **NumPy Documentation** – Used for numerical operations and data processing.
7. **Matplotlib Documentation** – Used for creating charts and visualizing analytical results.
8. **Seaborn Documentation** – Used for statistical data visualization and exploratory analysis.
9. **MySQL Documentation** – Used for SQL queries, aggregation, filtering, grouping, and business analysis.
10. **Microsoft Power BI Documentation** – Used for dashboard development, data visualization, filters, KPIs, and report creation.

Courses & Practice Platforms

11. **Google Data Analytics Professional Certificate – Coursera** – Referred to for data cleaning, analysis, visualization, and analytical methodology.
12. **HackerRank SQL** – Used for practicing SQL queries and improving database analysis skills.
13. **LeetCode SQL 50** – Used for practicing SQL problems and strengthening query-writing skills.
14. **Infosys Springboard – Python for Data Science** – Referred to for Python-based data analysis and data science concepts.

Project Resources

15. **Customer Transaction Dataset** – Used as the primary dataset for customer behavior, revenue, spending, segmentation, and churn analysis.
16. **Power BI Dashboard** – Developed as part of the project to present KPIs, trends, customer segments, product performance, and churn insights.

These references provided the required **technical knowledge, analytical methods, programming concepts, and visualization techniques** used throughout the project.

APPENDIX

The appendix contains the **supporting code, screenshots, graphs, SQL queries, and dashboard outputs** used in the development of the **Customer Behavior Analysis** project. These materials provide additional details about the practical implementation of the project and support the results discussed in the main report.

Appendix A – Dataset

The project uses a customer transaction dataset containing information related to **customers, purchases, products, payment methods, customer demographics, and churn**. The dataset was used for data cleaning, EDA, SQL analysis, customer segmentation, and Power BI visualization.

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	1.0	John Rivera	31	Female	0
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	1.0	John Rivera	31	Female	0
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	1.0	John Rivera	31	Female	0
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	0.0	John Rivera	31	Female	0
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	1.0	John Rivera	31	Female	0

Appendix B – Data Cleaning

Python and Pandas were used to inspect and prepare the dataset. The cleaning process included checking **data types, missing values, duplicate records, and date formats**.

```
#Data Cleaning
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(" ","_")
```

```
df.isnull().sum() •••
```

```
df['returns'].value_counts(dropna=False) •••
```

```
df['returns']=df['returns'].fillna('No') •••
```

```
df.isnull().sum() •••
```

```
df.duplicated().sum() •••
```

```
df['gender'].unique() •••
```

Appendix C – Feature Engineering

New features were created from the `purchase_date` column to support time-based analysis:

- `purchase_year`
- `purchase_month`

- `purchase_day`
- `purchase_quater`

These features were used for monthly, yearly, daily, and quarterly analysis.

feature Engineering

```
[649]: df['purchase_year'] = df['purchase_date'].dt.year
       df['purchase_month'] = df['purchase_date'].dt.month
       df['purchase_day'] = df['purchase_date'].dt.day
       df['purchase_quater'] = df['purchase_date'].dt.quarter
```

```
df •••
```

```
[653]: # age_group

       df["age_group"]=pd.cut(
           df['age'],
           bins=[18,25,35,45,65,100],
           labels=['18-25','26-35','36-45','46-60','60+']
         )
```

```
[655]: #spending level
       df['spending level']=pd.qcut(
           df['total_purchase_amount'],
           q=3,
           labels=['LOW','MEDIUM','HIGH']

       )
```

```
[657]: df['price_category'] = pd.cut(
           df['product_price'],
           bins=3,
           labels=['Low Price','Medium Price','High Price']
       )
```

```
[659]: df['quantity_category'] = pd.cut(
           df['quantity'],
           bins=[0,2,5,10],
           labels=['Low','Medium','High']
       )
```

Appendix D – Exploratory Data Analysis

EDA was performed using **Pandas, Matplotlib, and Seaborn** to analyze revenue, customers, orders, product categories, payment methods, and purchasing behavior.

Important results include approximately **₹681.35M total revenue, 49,661 customers, 250K orders, and ₹2,725.39 average purchase**.

```
# ==========================
# Key Performance Indicators
# ==========================

total_revenue = df['total_purchase_amount'].sum()
average_purchase = df['total_purchase_amount'].mean()
total_customers = df['customer_id'].nunique()
total_orders = len(df)
maximum_purchase = df['total_purchase_amount'].max()
minimum_purchase = df['total_purchase_amount'].min()

print(f"Total Revenue         : {total_revenue:,.2f}")
print(f"Average Purchase      : {average_purchase:.2f}")
print(f"Total Customers       : {total_customers:,}")
print(f"Total Orders          : {total_orders:,}")
print(f"Maximum Purchase      : {maximum_purchase:,.2f}")
print(f"Minimum Purchase      : {minimum_purchase:,.2f}")
```

```
Total Revenue         : 681,346,299.00
Average Purchase      : 2725.39
Total Customers       : 49,661
Total Orders          : 250,000
Maximum Purchase      : 5,350.00
Minimum Purchase      : 100.00
```

Appendix E – SQL Business Analysis

SQL was used to perform customer and business analysis, including:

- Customer and revenue analysis
- Product-wise revenue analysis
- Customer spending analysis
- Customer segmentation
- Churn analysis
- Aggregation and filtering of transaction data

```
--13. Revenue by Product Category
SELECT
    product_category,
    SUM(total_purchase_amount) AS revenue
FROM customer
GROUP BY product_category
ORDER BY revenue DESC;
```

Result:

	product_category text	revenue numeric
1	Home	171138916
2	Clothing	170716122
3	Electronics	170146025
4	Books	169345236

Appendix F – Customer Segmentation & Churn Analysis

Customer segmentation was used to understand differences in customer value and purchasing behavior. The project analyzed **four customer segments (0–3)** and different spending levels.

Churn analysis was performed using the **Churn = 0 and Churn = 1** classification. The dashboard shows an overall churn rate of approximately **20%**.

```
--2.Churn Percentage
SELECT
    churn,
    COUNT(*) AS total_customers,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM customer), 2) AS churn_percentage
FROM customer
GROUP BY churn;
```

Result:

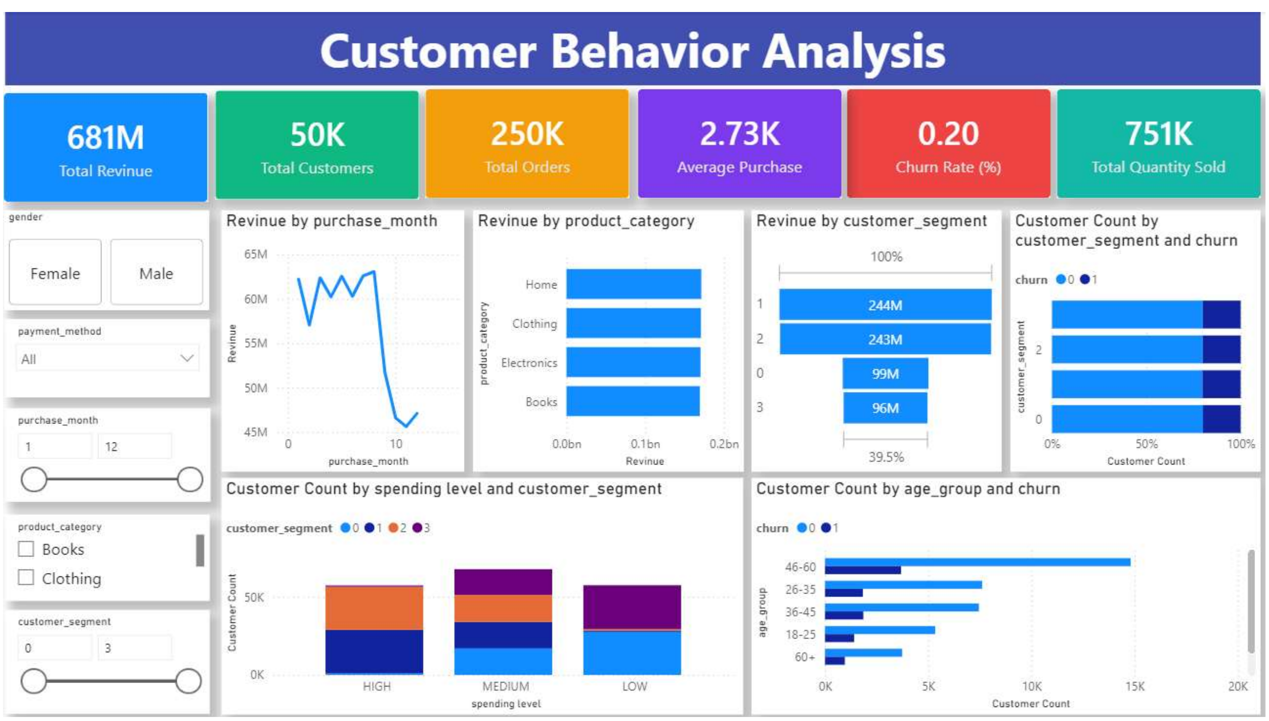
	churn bigint	total_customers bigint	churn_percentage numeric
1	0	199870	79.95
2	1	50130	20.05

Appendix G – Power BI Dashboard

The final Power BI dashboard presents the major project findings through **KPIs, charts, customer segments, spending levels, revenue trends, product analysis, and churn analysis**.

Key KPIs include:

- **Total Revenue:** ₹681.35M
- **Total Customers:** 49,661
- **Total Orders:** 250K
- **Average Purchase:** ₹2,725.39
- **Churn Rate:** 20%
- **Total Quantity:** 751K



Appendix H – Additional Graphs and Outputs

Additional graphs, tables, screenshots, code outputs, and analysis results generated during the project are included here for reference.


```
#You can describe the segments like this (adapt to your actual results):
#Segment Description of Above graph
#0  Young, low-spending customers
#1  High-value customers with high purchase amounts
#2  Medium-spending regular customers
#3  Older customers with high quantities purchased
```

Summary

The appendix provides supporting evidence for the **data preparation, analysis, SQL queries, segmentation, churn analysis, and Power BI dashboard** presented throughout this report.